

ANKUSH CHAUDHARY

FULL-STACK ENGINEER • AI ENGINEER

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PROFESSIONAL SUMMARY

Full-stack and AI engineer with 4+ years of production experience shipping web, mobile, and backend systems at scale, plus hands-on LLM engineering building agentic pipelines with LangGraph, LangChain, and the Anthropic API. At Tata Consultancy Services, owned a UK financial services platform serving 10K+ users, designed a reusable framework that cut project setup time by 35%, and was recognized with 8 performance awards. Recently completed an M.S. in Computer Engineering at Virginia Tech (GPA: 3.78/4.0).

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, SQL, C++
- **Frontend:** Angular, Ionic, React, HTML5, CSS3, UI/UX Development
- **Backend:** Spring Boot, Node.js, Flask, RESTful APIs, Microservices Architecture, JWT/Token Auth, Distributed Systems
- **Generative AI:** LangChain, LangGraph, Anthropic API, Prompt Engineering, RAG, Hugging Face
- **Cloud and DevOps:** AWS, Docker, Git, GitLab CI, GitHub Actions, Jenkins, CI/CD, Postman
- **Databases:** MySQL, MongoDB
- **Machine Learning:** NLP (BERT), Computer Vision (CNN), TensorFlow, PyTorch, Scikit-learn
- **Testing:** JUnit, Jest, Jasmine, Karma
- **Tools:** Jira, Figma, Agile/Scrum

PROFESSIONAL EXPERIENCE

AI Fellow | Handshake AI | *Remote* Oct 2025 - Nov 2025

- Authored adversarial prompts targeting model reasoning boundaries, successfully eliciting confidently incorrect responses on 5–10 high-difficulty DSA and computational mathematics problems.
- Stress-tested model performance on multi-step algorithmic reasoning problems including graph traversal, dynamic programming, and mathematical proof construction, exposing systematic failure patterns.
- Selected for an internal AI safety initiative focused on identifying and documenting failure modes in large language model responses under adversarial conditions.

Lead Software Engineering Intern | Shifting Waters Leadership Institute | *Richmond, VA* Jul 2025 - Sep 2025

- Served as sole technical owner of the organization's web platform, leading requirements gathering directly with the CEO, UI/UX design in Figma, full-stack development, and production deployment while mentoring one junior engineer to independent contribution.
- Built a responsive, mobile-first website implementing WCAG accessibility standards, SEO best practices, and Google Analytics tracking, delivering a production-ready platform within a 3-month engagement.
- Automated web form data ingestion into structured spreadsheets for executive reporting, building a repeatable extract-and-populate workflow that eliminated manual data entry for program staff across recurring service and program forms.

System Engineer | Tata Consultancy Services | *Pune, India* Nov 2020 - Jul 2024

- Built and shipped a production Angular and Ionic hybrid mobile application for a UK-based financial services client, serving 10K+ active users across a regulated environment with full end-to-end frontend ownership.
- Integrated 30+ RESTful API endpoints with JWT-based authentication flows, implementing route guards, HTTP interceptors, and session management to ensure secure, seamless access across all application surfaces.
- Automated CI/CD pipelines using GitLab across development and staging environments, accelerating deployment cycles by 40% and eliminating manual release steps across the full delivery workflow.
- Designed a reusable full-stack application framework covering project structure, authentication scaffolding, and base UI components, adopted by 2 additional client teams and reducing new project setup time by 35%.
- Led a frontend team of 4 engineers through final-year delivery, driving code quality through structured code reviews, pair programming, and onboarding documentation that accelerated new-member contribution.

- Conducted root cause analysis and debugging on production incidents by monitoring logs during releases, resolving issues in real time to minimize service disruption on a regulated financial platform.

EDUCATION

M.S. Computer Engineering | *Virginia Tech, Blacksburg, VA* | GPA: 3.78/4.0 Aug 2024 - May 2026

Coursework: Reinforcement Learning, Deep Learning, Machine Learning, Computer Vision, Information Security, Large-Scale Software Development, Brain-inspired Computer Architecture

B.E. Computer Science and Engineering | *SGB Amravati University, India* | CGPA: 8.89/10.0 Aug 2017 - Aug 2020

Coursework: Data Structures and Algorithms, Programming Languages, Operating Systems, Networking, Database SQL

TECHNICAL PROJECTS

Autonomous Job Search & Resume Pipeline | *Python, LangGraph, LangChain, Anthropic API, APScheduler, SerpAPI, Google Sheets API, LaTeX* May 2026

- Built an end-to-end autonomous job search system combining a multi-platform scraper (LinkedIn, Indeed, Google Jobs via SerpAPI) with a 6-node LangGraph agentic pipeline that automatically tailors resumes and cover letters for high-match roles.
- Engineered a cost-gated job-fit scoring system layering deterministic red-flag regexes, a content-hash JD cache, and a free-tier LLM rubric (Groq or local Ollama, with paid fallback) that filters roles by tech-stack fit, seniority, role type, and visa sponsorship signals, rejecting a bad-fit posting before any paid token is spent.
- Designed structured Pydantic output schemas, ATS keyword extraction, per-JD skill reordering, and a Jinja2-LaTeX PDF generation pipeline, reserving Claude Sonnet for final resume and cover-letter prose to keep per-application cost minimal while reducing tailoring time from hours to minutes.

Zero-Knowledge Password Authentication | *Circom, Cryptography, Node.js, React* | GitHub Jan 2025 - May 2025

- Built a secure authentication system using zero-knowledge proofs with Circom circuits and Poseidon hashing, verifying credentials without exposing passwords to the server.
- Developed a Node.js backend pipeline and React frontend for client-side proof generation, integrating end-to-end cryptographic testing to prevent phishing and replay attacks.

Efficient Multi-Task NLP with Task Vector Composition | *BERT, Machine Learning, NLP* | GitHub Mar 2025 - May 2025

- Developed a multi-task NLP model utilizing BERT and Task Vector Composition, achieving 87% accuracy on SST-2 and 74% F1 on NER.
- Maintained a fixed 110M parameter model size while dynamically scaling task-specific performance to reduce compute overhead.

Personal Budget Management System | *Python, Flask, PyQt6, MongoDB* | GitHub Aug 2025 - Nov 2025

- Engineered a full-stack desktop application utilizing a Flask REST backend, MongoDB persistence layer, and interactive PyQt6 UI for comprehensive finance tracking.
- Implemented MVC architecture with lazy-loaded database queries and optimized data-loading workflows to avoid unnecessary full-table reads on the finance-tracking UI.

HONORS & ACHIEVEMENTS

- Earned 8 TCS awards over 4 years, including Star of the Month, Star of the Quarter, Star Team, and On-the-Spot awards, for delivering scalable, production-ready software solutions.